Problem Set for 17 March 2025

Definition (Linear maps) Let f be a linear map. We restate the definition of image and pre-image (as mentioned in mathematical analysis, I believe) in formal and symmetric terms:

1. (the image) $f_*U = \{f(u) \mid u \in U\}$, where U is the subspace of the domain (domain: where f starts/maps from/originates);
2. (the inverse image) $f^*V = \{u \mid f(u) \in V\}$, where V is the subspace of the codomain (codomain: where f ends/maps to/terminates).

Exercise The most elementary part of linear algebra is to study the operations $\{f_*, f^*, +, \cap\}$. It is an enjoyable time to observe how these simple operations generate identities.

Prove the following (whenever we write $f_* X$,it is assumed that the linear space X is a subspace of the domain, and similarly for $f^* Y$):

1. $U \subset f^* f_* U$, when does equality hold for all U?
2. $f_* f^* V \subset V$, when does equality hold for all V?
3. $f_* f^* f_* = f_*$,
4. $f^* f_* f^* = f^*$,
5. $f_*(U+V) = f_* U + f_* V$;
6. $f^*(U \cap V) = f^* U \cap f^* V$;
7. Explain $f_*(U \cap V) \subset f_* U \cap f_* V$;
8. Explain $f^*(U+V) \supset f^* U + f^* V$;
9. **(optional)** if f is injective, then $f_*(U \cap V) = f_* U \cap f_* V$ always holds, does the converse hold?
10. **(optional)** if f is surjective, then $f^*(U+V) = f^* U + f^* V$ always holds, does the converse hold?
11. **(optional)** $f_*((f^* U) \cap V) = U \cap (f_* V)$;
12. **(optional)** $f^*((f_* U) + V) = U + (f^* V)$.

Definition

1. $\mathbb{F}[x] = \{\text{polynomials over } \mathbb{F}\} = \{\mathbb{F}\text{-sequences with finite non-zeros}\}$;
2. $\mathbb{F}[\![x]\!] = \{\text{formal power series over } \mathbb{F}\} = \{\mathbb{F}\text{-sequences}\}$.

Neither polynomials nor formal power series are functions in general:

1. for $\mathbb{F} = \{0, 1, +, \cdot\}$, the polynomials $x^2 + x$ and 0 represent the same functions;
2. for $\mathbb{F} = \mathbb{R}$, the formal power series $f(x) = \sum_{n\geq 1} x^n e^n$ diverges almost everywhere.

Problem This exercise (**without Axiom of Choice**) show that how $\mathbb{F}[x]$ and $\mathbb{F}[\![x]\!]$ differs.

1. Show that $\operatorname{Hom}_{\mathbb{F}}(\mathbb{F}[x], \mathbb{F}) \simeq \mathbb{F}[\![x]\!]$.

> *Hint: a linear map $\varphi \in \operatorname{Hom}_{\mathbb{F}}(\mathbb{F}[x], \mathbb{F})$ is uniquely determined by the image of basis, that is, $\{\varphi(x^k)\}_{k\geq 1}$.*

2. Show that $\operatorname{Hom}_{\mathbb{F}}(\mathbb{F}[\![x]\!], \mathbb{F})$ has a subspace which is isomorphic to $\mathbb{F}[x]$.
3. **(Optional)** What lies in $\operatorname{Hom}_{\mathbb{F}}(\mathbb{F}[\![x]\!], \mathbb{F}) \setminus \mathbb{F}[x]$?
4. **(Optional)** Show that $\mathbb{F}[\![x]\!]$ has a complementary subspace in $\operatorname{Hom}_{\mathbb{F}}(\operatorname{Hom}_{\mathbb{F}}(\mathbb{F}[\![x]\!], \mathbb{F}), \mathbb{F})$.
5. **(Optional)** Show that there are no surjective linear maps $\mathbb{F}[x] \to \mathbb{F}[\![x]\!]$.

> *Hint: there are surjective set maps from X to $\operatorname{Subsets}(X)$.*